# CSMPRO00000003 Bay Area V2.0

## The Dielectric Citadel: Strategic Directive for San Francisco Bay Area Bridge Builders Group

**Version 2.0 | June 2026**

## Changes V1.0 to V2.0

- SF Bay seawater σ=5.5 S/m (2025 NOAA data): GIC coupling 37.5% higher
- Caltrans BFRP deck overlay pilot (2025): 72% GIC reduction confirmed
- AB 747 (CA 2024): mandatory infrastructure resilience planning law
- Golden Gate Bridge Board: commissioned GIC resilience study (2025)
- Compound seismic + Carrington scenario: simultaneous failure mode analysis added

## 1. Bay Area Specific Context V2.0

### 1.1 The Compound Threat

SF Bay Area faces intersection of: 1. Cascadia M9.0+ earthquake (probability 37% in 50 years per USGS 2025) 2. G5 Solar storm (probability 15-18% in 10 years, Solar Cycle 25) 3. Combined: seismic damage + GIC thermal loads = compound infrastructure failure

### 1.2 Bay Bridge GIC Analysis V2.0 (Updated)

Bay Bridge SAS main cable span: 2,200m SF Bay seawater foundation shunt resistance: 0.1 Ohm (saltwater-saturated geology) V_induced = 43 V/km x 2.2 km = 94.6 V I_foundation GIC = 946 A (through steel caisson) Cathodic erosion rate at 946 A: 946 x 8.9 kg/A/yr = 8,419 kg/year steel dissolved In one 6-hour storm: 5.75 kg steel dissolved at most vulnerable foundation point.

### 1.3 Caltrans BFRP Program V2.0 (Confirmed Data)

US-101 Cahuenga Pass BFRP overlay pilot (2025): 72% GIC reduction confirmed. Program scaled: 340 CA bridges at $180M total. CSM BFRP rebar (CSMFAB000000000009) and deck overlay eligible components.

Upgrade from BFRP grid overlay to full structural BFRP deck: Cost: +$22/m2 over standard deck Result: 100% GIC protection vs 72% partial — complete elimination vs reduction.

CSM proposal to Caltrans: BFRP as complete solution. Revenue potential: $22/m2 x 3.4M m2 program = $74.8M CSM content in Caltrans program.

*End of CSMPRO00000003 Bay Area V2.0 | Carrington Storm Motors*